

Gemini Chat

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [123, 110, 97, 104, 92, 118, 101, 97, 92, 99]

Tap amplitudes (pixels): [129.65338406690356, 129.5414991421668, 115.83177456984762, 110.01818031580054, 116.84605256490268, 107.33592129385204, 95.01578816175763, 116.76472069936193, 116.8289347721702, 114.12712210513327, 116.81181447096864]



Based on the extracted features, here is the analysis and scoring for MDS-UPDRS Item 3.4.

Analysis

- **Rhythm & Speed (Interruptions/Slowing):**
 - **Data:** Tap periods range from 92 to 123 frames.
 - **Observation:** The rhythm is generally stable. There are no extreme outliers that would indicate a "halt," "freeze," or significant interruption (e.g., a period suddenly jumping to 300+ frames). The variation (92–123) is within a normal range of variance for human movement.
 - **Conclusion:** No significant hesitations or slowing detected.
- **Amplitude (Decrement):**
 - **Data:** [129.7, 129.5, 115.8, 110.0, 116.8, 107.3, 95.0, 116.8, ...]
 - **Observation:**
 - **Taps 1-2:** High amplitude (~129).
 - **Tap 3:** Distinct drop to ~115.

- **Tap 4-11:** Amplitude fluctuates but generally stays in the 110–116 range (with a dip at Tap 7).
- **Conclusion:** There is a clear decrement in amplitude. It does not occur "near the end" (Score 1 criteria). It occurs early (after the 2nd tap). However, it does not progressively decline to zero; it stabilizes at a slightly lower level.

Scoring Logic

- **Score 0 (Normal):** Incorrect. The amplitude drop is noticeable.
- **Score 1 (Slight):** Incorrect. The decrement happens early (Tap 3), not near the end of the sequence.
- **Score 3 (Moderate):** Unlikely. This requires decrement starting *after the 1st tap* or severe interruptions. Here, the first two taps are consistent (129.6 vs 129.5), and the decrement starts after the 2nd tap. Additionally, the rhythm is preserved.
- **Score 2 (Mild): Correct.** The criteria includes "mild slowing" or "amplitude decrements midway." While the decrement is earlier than "midway," the overall performance—maintained rhythm, no freezing, and valid amplitude maintenance after the initial drop—best fits the "Mild" classification. The drop is distinct but the subject recovers and finishes the task successfully.